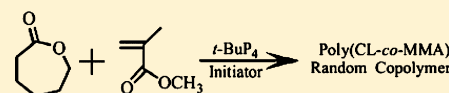


Hybrid Copolymerization of ϵ -Caprolactone and Methyl MethacrylateHongjun Yang,[†] Jinbao Xu,[‡] Stergios Pispas,^{*,§} and Guangzhao Zhang^{*,†,‡}[†]Hefei National Laboratory for Physical Sciences at Microscale, Department of Chemical Physics, University of Science and Technology of China, Hefei 230026, P. R. China[‡]Faculty of Materials Science and Engineering, South China University of Technology, Guangzhou 510640, P. R. China[§]Theoretical and Physical Chemistry Institute, National Hellenic Research Foundation, 48 Vassileos Constantinou Avenue, 11635 Athens, Greece

S Supporting Information

ABSTRACT: Copolymerization of ϵ -caprolactone and methyl methacrylate with (1-*tert*-butyl-4,4,4-tris(dimethylamino)-2,2-bis[tris(dimethylamino)-phosphoranylideneamino]-2 Λ 5,4 Λ 5-catenadi(phosphazene) (*t*-BuP₄) as the catalyst has been studied. Nuclear magnetic resonance (NMR) and differential scanning calorimetry (DSC) measurements indicate that the cyclic ester and vinyl monomer form a random copolymer. Thermogravimetric analysis (TGA) measurements demonstrate that such a random copolymer exhibits a decomposition temperature higher than those of the corresponding homopolymers. The present study reveals a “hybrid copolymerization” i.e., a combination of vinyl polymerization and ring-opening polymerization, which makes the incorporation of cyclic and vinyl monomers into the same polymer chain possible. Hybrid copolymerization is expected to produce a number of new interesting polymers.



■ INTRODUCTION

Copolymerization of cyclic monomers with vinyl monomers is very interesting because it can produce new types of copolymers and open up new research directions in the fields of polymer chemistry and of polymer applications. However, it is very difficult because the reactivity and polymerization mechanism of the two types of monomers are extremely different. So far, only the copolymerization of *N*-phenylmaleimide (electrophilic) with some nucleophilic cyclic monomers via zwitterion was reported.^{1–4} For common vinyl monomers and cyclic monomers, the copolymerization has not been realized yet.

(1-*tert*-Butyl-4,4,4-tris(dimethylamino)-2,2-bis[tris(dimethylamino)-phosphoranylideneamino]-2 Λ 5,4 Λ 5-catenadi(phosphazene) (*t*-BuP₄) can independently catalyze ring-opening polymerization (ROP) of cyclic esters and anionic addition polymerization of vinyl monomers with alcohol or ethyl acetate (EA) as the initiator.^{5–15} In the present work, we report that *t*-BuP₄ can catalyze the copolymerization of ϵ -caprolactone (CL) with methyl methacrylate (MMA), yielding a random copolymer.

■ EXPERIMENTAL SECTION

Materials. ϵ -Caprolactone (CL) from Aldrich was dried over calcium hydride (CaH₂) and distilled under reduced pressure prior to use. Ethylene glycol (EG) and methanol (MeOH) from Aldrich were dried over sodium with nitrogen protective atmosphere and distilled under vacuum after refluxing for hours. Ethyl acetate (EA) from Sinopharm was washed with aqueous 5% Na₂CO₃, dried with K₂CO₃, and then distilled from CaH₂. Methyl methacrylate (MMA) from Sinopharm was distilled from CaH₂ and stored under a nitrogen

atmosphere at -20 °C. Tetrahydrofuran (THF) and toluene (TOL) from Sinopharm were freshly distilled from sodium/benzophenone and stored under an argon atmosphere. *t*-BuP₄ and 1-naphthalene-methanol (NMA) from Aldrich were used as received. Other reagents from Sinopharm were used as received.

Characterization. *Nuclear Magnetic Resonance.* ¹H NMR, ¹³C NMR, and 2D NMR spectra (¹H–¹H COSY and ¹H–¹³C HMBN NMR spectra) were recorded on a Bruker AV400 NMR spectrometer by using deuterated chloroform (CDCl₃) as solvent and tetramethylsilane (TMS) as internal standard.

Size Exclusion Chromatography (SEC). The molecular weight and polydispersity index (PDI) were measured at 35 °C on a Waters SEC equipped with a model 510 pump, a differential refractive index detector model 410 (RI), and a Waters 2487 UV detector working at a wavelength of 254 nm. A series of monodisperse polystyrenes were used as standards. THF was used as the eluent at a flow rate of 1.0 mL/min.

Thermal Analysis. Thermogravimetric analysis (TGA) was performed on a TA SDT Q600 instrument under nitrogen atmosphere at a heating rate of 10 °C/min in the range 25–500 °C. Differential scanning calorimetry (DSC) was conducted on a TA Instruments Q2000 under a nitrogen flow of 50 mL/min. Samples were quickly heated to 150 °C and kept for 10 min to remove thermal history and then cooled to -80 °C at a rate of 10 °C/min. Finally, they were reheated to 150 °C at the same rate.

Polymerization of CL in the presence of t-BuP₄. A typical polymerization was performed as follows. CL (0.46 g, 4 mmol, 100 equiv), EA (3.90 μ L, 0.04 mmol, 1.0 equiv), and toluene (0.5 mL) were placed in a flamed and nitrogen purged round-bottom flask equipped with a magnetic stirrer. *t*-BuP₄ (40 μ L, 0.04 mmol, 1.0 equiv

Received: February 11, 2012

Revised: March 29, 2012

Published: April 12, 2012



Table 1. Characterization Data of the Polymers^a

sample	f^b (mol/mol)	initiator	$M_n \times 10^{-4}$ (g/mol)	M_w/M_n	f_p^c (mol %)	conv.-CL (mol %)	conv.-MMA (mol %)
S1	0/100	MeOH	2.34	1.96	100		100
S2	25/75	MeOH	2.05	2.32	76.1	100	99.0
S3	50/50	MeOH	2.82	1.72	55.9	100	100
S4	75/25	MeOH	1.61	1.54	29.0	97.2	99.3
S5	100/0	MeOH	1.91	1.45	0	100	
S6	100/0	EA	1.87	1.50	0	100	
S7	50/50	MeOH	1.47	1.84	54.8	99.0	100
S8	50/50	MeOH	1.93	1.71	53.2	100	100
S9	50/50	EG	3.14	4.36	49.6	99.5	98.1
S10	50/50	EA	2.56	2.26	62.3	100	100
S11	25/25	MeOH	0.81	1.44	54.5	100	100
S12	100/100	MeOH	4.13	1.67	50.2	99.6	100
S13	50/50	NMA	4.73	2.01	49.6	100	100
S14 ^d	10/10	MeOH	0.18	2.87	54.2	87.4	76.8
F1 ^e	50/50	MeOH	2.14	1.39	55.7		
F2 ^f	50/50	MeOH	5.06	1.23	56.1		

^aMolar ratio of monomer to initiator is 100/1 except samples S11 (50/1), sample S12 (200/1) and sample S14 (20/1), reaction temperature is 25 °C except that in S8 (−25 °C), solvent is THF except that in S7 (toluene). ^bMolar ratio of CL to MMA. ^cMolar fraction of MMA units in the copolymer determined by ¹H NMR. ^dThe reaction time is 2 min. ^eThe low molecular weight fraction from sample S3. ^fThe high molecular weight fraction from sample S3.

in hexane) was added through a rubber septum with a syringe to start the polymerization at 25 °C. The reaction was terminated with hydrochloric acid/methanol (1/20 v/v) after one hour. The product was dissolved in CH₂Cl₂ and precipitated into a large excess of cold methanol. The polymer was dried under vacuum. Polymerizations of CL or MMA initiated by MeOH were carried out using a similar procedure.

Copolymerization of CL and MMA. A typical copolymerization was performed as follows. CL (0.23 g, 2 mmol, 50 equiv), MMA (0.20 g, 2 mmol, 50 equiv), MeOH (2.24 μL, 0.04 mmol, 1.0 equiv), and toluene (0.5 mL) were placed in a flamed and nitrogen purged round-bottom flask equipped with a magnetic stirrer. The flask was placed in an oil bath at 25 °C. *t*-BuP₄ (40 μL, 0.04 mmol, 1.0 equiv in hexane) was added through a rubber septum with a syringe to start the polymerization. The polymerization was completed in 12 h and was terminated with hydrochloric acid/methanol (1/20 v/v). The product was dissolved in THF and precipitated into a large excess of cold methanol. The polymer was dried under vacuum. Copolymerization of CL and MMA under other reaction conditions was carried out using a similar procedure, and the results are summarized in Table 1.

RESULTS AND DISCUSSION

Copolymerization of CL and MMA was first investigated in THF at room temperature with MeOH as the initiator. The molar ratio of [MMA]/[CL]/[*t*-BuP₄]/[MeOH] was 50/50/1/1. Figure 1 shows the typical ¹H NMR spectra of the resulting copolymer. The signals at 1.25–1.47, 1.51–1.71, 2.18–2.40, and 3.96–4.17 ppm are due to the protons in CL units. The signals at 0.72–1.16, 1.72–2.14, and 3.51–3.71 ppm are assigned to MMA units. Note that for PMMA homopolymer, a singlet peak at around 3.5–3.7 ppm can be observed for the methoxy proton (–OCH₃). However, the peak corresponding to the methoxy proton of MMA unit for the copolymer shows splitting, indicating the presence of the comonomeric sequence with different configurations or the formation of CL/MMA copolymer.¹⁶ In addition, the molar fraction (f_p) of MMA in the copolymer determined by ¹H NMR was approximately equal to the feed ratio (f) (Table 1). To further verify the copolymerization, we synthesized a low molecular weight sample (S14) with degree of polymerization (DP) of 17, where the polymerization was terminated with hydrochloric acid/

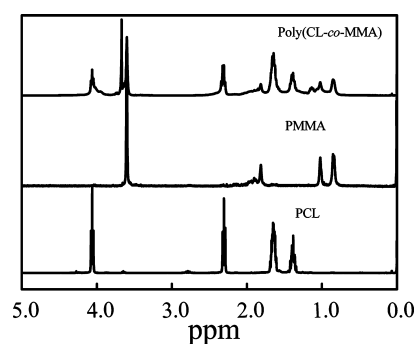


Figure 1. ¹H NMR spectra for poly(CL-*co*-MMA) copolymer (sample S3), PCL (sample S5), and PMMA (sample S1).

methanol (1/20 v/v) within two minutes. The ¹H NMR spectrum (Figure S1 in Supporting Information) of the sample is similar to that of the sample (S3) with DP of 100, indicating that copolymerization is rapid and occurs at the initial stage of the allowed polymerization time. This is also supported by the high degree of conversion for both monomers at the short polymerization time.

Figure 2 shows the ¹³C NMR spectra for poly(CL-*co*-MMA) copolymer. In comparison with the spectra of PCL and PMMA homopolymers (see Figures S2 and S3 in Supporting Information), we can observe three new peaks. The peaks at 33.97–33.9 and 51.4–51.5 ppm are assigned to the quaternary (–CH₂C(CH₃)(COOCH₃)CO–) and secondary carbons (–OCH₂C(CH₃)(COOCH₃)–) in MMA unit in the proximity of the CL unit. The peak at 173.9–174.0 ppm is due to the ester carbonyl (–C(CH₃)(COOCH₃)CO–) of CL unit adjacent to a MMA unit. Clearly, CL and MMA copolymerize into a random copolymer (poly(CL-*co*-MMA)) with mixed triad sequences.

Figure 3 shows the two-dimensional ¹H–¹³C HMBC NMR spectra of a poly(CL-*co*-MMA) copolymer. Clearly, the protons of –CH₃ in MMA units at 0.78–1.01 ppm are related to that of the –OCH₂ carbon atom, indicating the presence of CL-MMA sequences in the copolymer chain. On the other hand, the

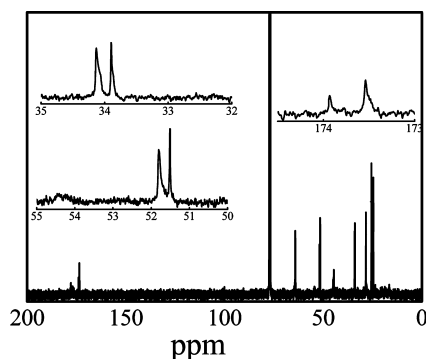


Figure 2. ^{13}C NMR spectra for poly(CL-co-MMA) copolymer (sample S3).

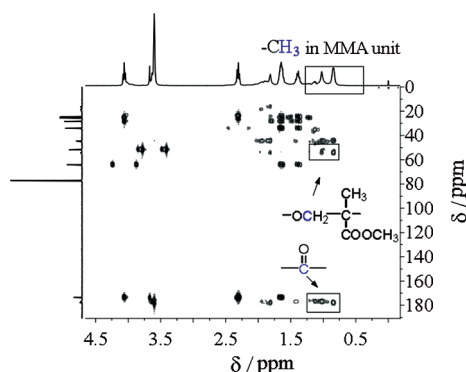


Figure 3. ^1H – ^{13}C HMBC NMR spectra for the copolymers initiated by MeOH (sample S3).

protons of $-\text{CH}_3$ in MMA units have a correlation with the ester carbonyl atom in CL units at 173.9–174.0 ppm, indicating that the copolymer chain has MMA-CL sequences. These observations further demonstrate that the derived poly(CL-co-MMA) is a random copolymer.

Figure 4 shows SEC curves for the resultant polymer (RI detection). The SEC-RI curve displays a bimodal distribution,

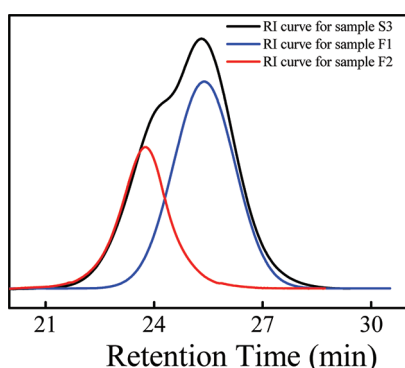


Figure 4. SEC curves for the poly(CL-co-MMA) copolymer before and after fractional precipitation (sample S3).

suggesting the presence of two components with quite different molecular weights. The two components (samples F1 and F2) were separated by fractional precipitation. It was found, by NMR, that they have similar compositions (f_p), although they possess different molecular weights (Table 1). Accordingly, the bimodal molecular weight distribution may be due to the nucleophile attack on carbonyl group of MMA, which often

happens during anionic polymerization of MMA.^{17–19} However, probably because of their low content, signals related to such bonds were not detected in NMR spectra.

Figure 5 shows the DSC curves for the resulting copolymers. All the copolymers exhibit only one glass transition. Sample S4

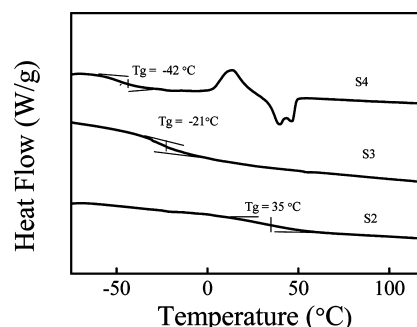


Figure 5. DSC curves for the poly(CL-co-MMA) copolymers (S2–S4) at a heating rate of 10 °C/min.

with higher CL content also shows a melting point, presumably due to crystallization of CL segments. The glass transition temperature (T_g) is between those of poly(methyl methacrylate) (PMMA) ($T_g \sim 115$ °C) and poly(ϵ -caprolactone) (PCL) ($T_g \sim -60$ °C) homopolymers (see Figures S4 and S5 in Supporting Information), and T_g decreases with CL content in the copolymer. Particularly, the T_g values agree with those predicted by the Fox equation.²⁰ Similar results were obtained at the scanning rate of 5 °C/min. Thus, the NMR and DSC results indicate that CL and MMA segments are randomly distributed along the polymer chain. In addition, when CL content is low enough (e.g., for samples S2 and S3), the melting endotherm specific to PCL is no longer observed, further indicating the random incorporation of MMA and CL units. In the case of sample S4, CL sequences are long enough for the formation of crystallites.

Figure 6 shows the TGA curves of poly(CL-co-MMA), together with those for PMMA and PCL homopolymers. PCL

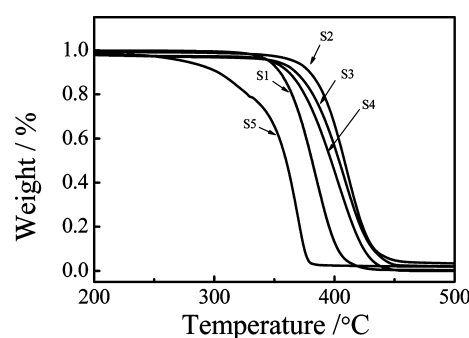


Figure 6. TGA curves for poly(CL-co-MMA) (samples S2 to S4), PMMA (sample S1), and PCL (sample S5) homopolymers.

starts to degrade at about 250 °C in a nitrogen atmosphere, and degradation is completed at about 380 °C. For PMMA, the degradation begins at about 330 °C and ends at about 420 °C. All copolymers exhibit improved thermostability in comparison with the two homopolymers, and the thermal decomposition temperature increases with MMA content in the range of f_p from 29.0 mol % to 76.1 mol %. This should be associated with the chemical connection of the CL and MMA units in the

copolymer. Anyhow, this fact further indicates that CL and MMA form random copolymers under the applied polymerization conditions.

We also investigated the effect of different initiators (MeOH, EA, and EG) on the copolymerization (Table 1). Each SEC curve for the copolymer in Figure 7 shows multiple peaks,

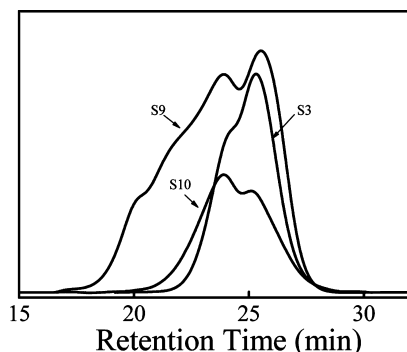


Figure 7. SEC curves for the copolymers initiated by MeOH (sample S3), EA (sample S10), and EG (sample S9), respectively.

indicating several species with different molecular weights. The polymerization with EA or MeOH as the initiator makes a little difference in the molecular weight and molecular weight distribution of the copolymer. However, the initiation by EG (a potential difunctional initiator) results in a higher molecular weight ($M_n = 3.14 \times 10^4$ g/mol) and a broader PDI ($M_w/M_n = 4.36$). This is probably because the chain grows from the two active centers in EG.²¹

To gain some insight into initiator efficiency, 1-naphthalenemethanol with a UV-active aromatic moiety was also used to initiate the copolymerization. Figure 8 shows SEC curves for

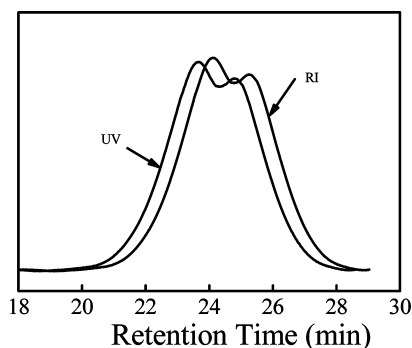


Figure 8. SEC curves for the copolymer initiated by 1-naphthalenemethanol by RI and UV detection, respectively.

the resultant polymer (RI and UV detection). Either the SEC trace monitored with RI detector or that obtained with UV detector has a bimodal distribution, suggesting two components with quite different molecular weights. More importantly, they are similar in shape and are characterized by the same retention time, indicating that the two components should have similar chemical structures,²² and that the initiator group is incorporated onto the chain end without chain scission taking place.

To ensure that *t*-BuP₄ acts as the organocatalyst, the polymerization without initiator has been examined. The yield was found to be only 7.2%. The limited polymerization observed in this case may be initiated by trace amounts of active

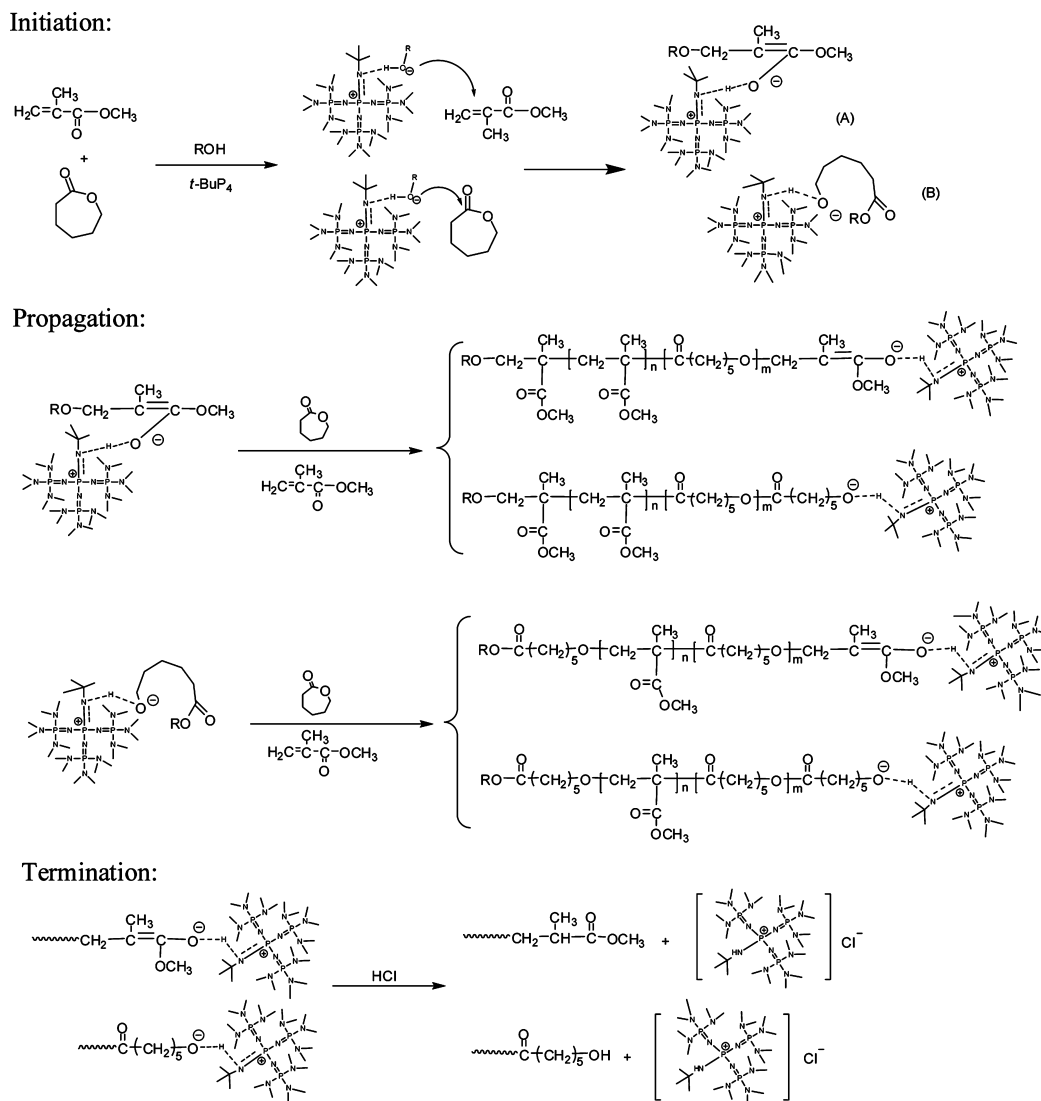
hydrogens present in the monomers or *t*-BuP₄ solution.²³ The experimental results demonstrate that *t*-BuP₄ acts as a catalyst in the particular copolymerization. In addition, a number of experiments carried out in parallel show that NaOMe cannot initiate the CL/MMA copolymerization. Thus, a simple deprotonation mechanism under the influence of *t*-BuP₄ should not be operative in this copolymerization system.

The polymerization was also conducted in toluene, and no significant effects were observed by changing the solvent. Polymerization at a temperature as low as -25 °C shows that temperature has little effect on the polymerization outcome (Table 1). The results summarized in Table 1 also show that the molecular weight distribution (M_w/M_n) of the copolymer increases as the molar ratio of MMA increases in the feed. This is probably because the propagating nucleophile adds to the carbonyl of MMA segments.^{17–19}

On the basis of the results at hand, we suggest that the copolymerization of CL and MMA catalyzed by *t*-BuP₄ involves the following steps (Scheme 1). (1) Initiation. This step follows a hydrogen bonding mechanism.^{24–27} Phosphazene base (*t*-BuP₄) first activates the alcohol by the intermolecular hydrogen bonding, and the nucleophilic attack on the monomer produces two kinds of active centers (A and B). (2) Propagation. Both of the active centers react with either MMA or CL to form new active centers, and the reactive intermediate is continuously regenerated by addition of monomers leading to a long active chain. (3) Termination. The propagating chain is deactivated by the addition of hydrochloric acid.

The scheme shown above does not represent all details during polymerization but depicts some general aspects of the polymerization process. The mechanism of the copolymerization seems complex, and some parallel reactions/transformations and side reactions may happen. For example, the nucleophilic active centers may add onto the carbonyl of MMA segments,^{17–19} leading to a broader molecular weight distribution. Isomerization of hydrogen bonded stabilized “pseudo-enolate” type ions (where the negative charge is located mostly on the O atom) to vinyl type carbanions (where the negative charge is located on the terminal vinylic C atom) may take place so that the monomers are incorporated into the copolymer main chain via addition to the C=C bonds. Multicenter interactions involving the dimethylamino groups of *t*-BuP₄ are also possible. Chain transfer and chain termination in other ways could occur, e.g., reaction of active centers with CL carbonyl or complete abstraction of H from protonated phosphazene by reaction with a MMA type active chain end carbanion, leading to the regeneration of the free base. More studies are needed to fully elucidate all aspects of the reaction mechanism which, however, are beyond the scope of the present manuscript but will be the subject of future work. It should be noted that this copolymerization is quite different from that taking place via zwitterions, where the electrophilic and nucleophilic monomers react to generate a zwitterionic intermediate and the resulting copolymers usually have alternating structures.^{28,29} In the present copolymerization, each monomer can form propagating species, which can further combine with either similar or dissimilar monomers, forming a random copolymer. Therefore, we propose the term “hybrid polymerization” for this type of copolymerization between cyclic and vinyl monomers via organocatalysis with *t*-BuP₄ base.

Scheme 1. Possible Mechanism for CL/MMA Hybrid Copolymerization



CONCLUSIONS

In conclusion, copolymerization of CL with MMA is possible and can be carried out using *t*-BuP₄ as the organocatalyst. Characterization data demonstrate that the resulting copolymer has a random structure and exhibits a decomposition temperature higher than those of the homopolymers. The hybrid polymerization presented here is applicable to a common cyclic ester and a vinyl monomer. Thus, a number of new polymers with interesting properties can be produced by this polymerization. Studies are in progress that are aiming at a better control of the polymerization reaction in order to produce more well-defined polymers, i.e., by the use of different phosphazene bases. The elucidation of the mechanism of the hybrid copolymerization is another future goal.

ASSOCIATED CONTENT

Supporting Information

DSC curves and NMR spectra of some samples. This material is available free of charge via the Internet at <http://pubs.acs.org>.

AUTHOR INFORMATION

Corresponding Author

*Phone: +86-20-22236373. Fax: +86-20-22236373. E-mail: msgzzhang@scut.edu.cn.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The financial support of National Program on Key Basic Research Project (2012CB933802), the National Distinguished Young Investigator Fund (20725414), and the Fundamental Research Funds for the Central Universities is acknowledged.

REFERENCES

- (1) Simionescu, C. I.; Grigoras, M.; Bicu, E.; Onofrei, G. *Polym. Bull.* **1985**, *14*, 79–83.
- (2) Rivas, B. L.; del C. Pizarro, G.; Canessa, G. S. *Polym. Bull.* **1988**, *19*, 123–128.
- (3) Hagiwara, T.; Takeda, M.; Hamana, H.; Narita, T. *Macromolecules* **1989**, *22*, 2025–2026.
- (4) Ikeda, Y.; Yoshida, Y.; Ishihara, K. *Macromol. Rapid Commun.* **1996**, *17*, 713–721.

- (5) Zhang, L.; Nederberg, F.; Pratt, R. C.; Waymouth, R. M.; Hedrick, J. L.; Wade, C. G. *Macromolecules* **2007**, *40*, 4154–4158.
- (6) Winter, J. D.; Coulembier, O.; Gerbaux, P.; Dubois, P. *Macromolecules* **2010**, *43*, 10291–10296.
- (7) Pietzonka, T.; Seebach, D. *Angew. Chem., Int. Ed.* **1993**, *32*, 716–717.
- (8) Makiguchi, K.; Satoh, T.; Kakuchi, T. *Macromolecules* **2011**, *44*, 1999–2005.
- (9) Zhang, L.; Pratt, R. C.; Nederberg, F.; Horn, H. W.; Rice, J. E.; Waymouth, R. M.; Wade, C. G.; Hedrick, J. L. *Macromolecules* **2010**, *43*, 1660–1664.
- (10) Boileau, S.; Illy, N. *Prog. Polym. Sci.* **2011**, *36*, 1132–1151.
- (11) Nederberg, F.; Connor, E. F.; Müller, M.; Glauser, T.; Hedrick, J. L. *Angew. Chem., Int. Ed.* **2001**, *40*, 2712–2715.
- (12) Lohmeijer, B. G. G.; Pratt, R. C.; Leibfarth, F.; Logan, J. W.; Long, D. A.; Dove, A. P.; Nederberg, F.; Choi, J.-S.; Wade, C.; Waymouth, R. M.; Hedrick, J. L. *Macromolecules* **2006**, *39*, 8574–8583.
- (13) Pratt, R. C.; Lohmeijer, B. G. G.; Long, D. A.; Waymouth, R. M.; Hedrick, J. L. *J. Am. Chem. Soc.* **2006**, *128*, 4556–4557.
- (14) Schmalz, H.; Lanzendörfer, M. G.; Abetz, V.; Müller, A. H. E. *Macromol. Chem. Phys.* **2003**, *204*, 1056–1071.
- (15) Kakuchi, T.; Chen, Y.; Kitakado, J.; Mori, K.; Fuchise, K.; Satoh, T. *Macromolecules* **2011**, *44*, 4641–4647.
- (16) Grabe, N.; Zhang, Y.; Agarwal, S. *Macromol. Chem. Phys.* **2011**, *212*, 1327–1334.
- (17) Bhattacharya, A.; Misra, B. N. *Prog. Polym. Sci.* **2004**, *29*, 767–814.
- (18) Hong, K. L.; Uhrig, D.; Mays, J. W. *Curr. Opin. Solid State Mater. Sci.* **1999**, *4*, 531.
- (19) Allen, P. E. M.; Casey, B. A.; Dankiw, W. J. *Macromol. Sci., Part A: Pure Appl. Chem.* **1970**, *4*, 1091–1103.
- (20) Fox, T. G. *Bull. Am. Phys. Soc.* **1956**, *1*, 123–124.
- (21) Warzelhan, V.; Höcker, H.; Schulz, G. V. *Makromol. Chem.* **1980**, *181*, 149–163.
- (22) Laguna, M. T. R.; Gallego, J.; Mendicuti, F.; Saiz, E.; Tarazona, M. P. *Macromolecules* **2002**, *35*, 7782–7790.
- (23) Zhao, J. P.; Zhang, G. Z.; Pispas, S. J. *Polym. Sci., Part A: Polym. Chem.* **2011**, *48*, 2320–2328.
- (24) Mori, S.; Suzuki, T. *J. Liq. Chromatogr.* **1981**, *4*, 1685–1696.
- (25) Zhang, L.; Nederberg, F.; Messman, J. M.; Pratt, R. C.; Hedrick, J. L.; Wade, C. G. *J. Am. Chem. Soc.* **2007**, *129*, 12610–12611.
- (26) Kamber, N. E.; Jeong, W.; Waymouth, R. M.; Pratt, R. C.; Lohmeijer, B. G. G.; Hedrick, J. L. *Chem. Rev.* **2007**, *107*, 5813–5840.
- (27) Kiesewetter, M. K.; Shin, E. J.; Hedrick, J. L.; Waymouth, R. M. *Macromolecules* **2010**, *43*, 2093–2107.
- (28) Saegusa, T.; Kobayashi, T.; Kimura, Y.; Chow, T.; Kobayashi, S. *Macromolecules* **1979**, *12*, 533–535.
- (29) Saegusa, T.; Kimura, Y.; Ishikawa, N.; Kobayashi, S. *Macromolecules* **1976**, *9*, 724–727.